IM235AI

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

III Semester B. E. Regular / Supplementary Examinations Feb/Mar-2025

Industrial Engineering and Management

DIGITAL METROLOGY

lours *Maximum Marks: 100*

s to candidates:

swer all questions from Part A. Part A questions should be answered in first three ges of the answer book only.

swer FIVE full questions from Part B. In Part B question number 2 is compulsory.

swer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

		M	BT	CO
1	Differentiate between Static and Dynamic Characteristics of a measurement system.	02	2	1
2	State the importance of calibration in a measurement system.	02	1	1
3	List any four key specifications of a sensor.	02	1	2
4	Mention the main components of a data acquisition system.	02	1	2
5	Define tolerance.	02	1	3
.6	Highlight any two functions of optical flats.	02	2	3
.7	What is meant by Fit?	02	1	3
.8	What is the Root Mean Square (*RMS*) value in surface finish measurement?	02	1	3
.9	Mention the type of *CMM* which is best suited for inspecting large and heavy workpieces.	02	2	4
.10	State the primary use of a laser in laser metrology.	02	2	4

PART-B

ı	Explain the elements of a generalized measurement system with a neat sketch.	08	2	1
ɔ	Explain the importance of calibration in measurement systems.	08	2	1
a	Explain the working of a potentiometer as a displacement sensor.	08	3	2
ɔ	State and discuss the working principle of a tacho generator.	08	3	2
	OR			
a	Discuss the working of piezoelectric sensors for pressure measurement.	08	3	2
b	Explain the elements of a data acquisition system and the role of *A/D* and *D/A* converters.	08	2	2
a	With suitable examples, differentiate between interchangeability and selective assembly.	08	3	3
b	Differentiate between the hole basis system and the shaft basis system. Identify which is more commonly preferred, and provide the fits designation as per *IS* 919 – 1963.	08	3	3
	OR			

6	a	Explain the significance of tolerance in manufacturing. Discuss the various types of tolerances.	
	b	What is Geometric Dimensioning and Tolerancing (*GD&T*)? And elaborate on the concept of position tolerance.	
7	a	Explain the gear tooth Vernier method used to measure the thickness of gear teeth.	
	b	Discuss the application of optical flats in interferometry for flatness measurement.	
		OR	
8	a	With the help of a neat sketch, explain the working and applications of a Tool Maker's Microscope.	
	b	Elaborate on the use of Tomlinson Surface Meter in measuring surface finish.	
9	a	Describe the constructional features and working principle of a Coordinate Measuring Machine.	
	b	Explain the role of digital devices in computer-aided inspection with a Coordinate Measuring Machine.	
		OR	
10	a	Identify the different types of Coordinate Measuring Machine (CMMs) and their specific applications.	
	b	Mention the principle and applications of the Michelson Interferometer in laser metrology.	

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III Semester B. E. Examinations Apr-2024

Industrial Engineering and Management

DIGITAL METROLOGY

3 Hours *Maximum Marks: 100*

tions to candidates:

Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

Answer FIVE full questions from Part B. In Part B question number 2 is compulsory.

Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

		M	BT	CO
1.1	Discuss the differences between static and dynamic characteristics of a measurement system.	02	1	1
1.2	List the Errors in measurements.	02	2	2
1.3	Explain the functioning of a Piezoelectric Sensor and provide example for its applications.	02	2	2
1.4	List the applications of a Capacitive Transducer in Proximity Sensing.	02	2	2
1.5	What is meant by "Basic Hole" and "Basic Shaft"?	02	2	3
1.6	How will you write the fit: Shaft $'m'$ of grade 5 and hole $'H'$ of grade 11 and the basic size of $40mm$?	02	1	3
1.7	An optical flat can be employed to measure height differences in which range?	02	2	2
1.8	Describe the working principle of an Autocollimator.	02	1	2
1.9	Describe the working principle of a Michelson Interferometer.	02	2	4
1.10	List four applications of Laser Metrology in Industry.	02	2	4

PART-B

	Enumerate on the generalized measurement system. Justify the answer with an example, neat sketches and block diagrams.	08	3	1
	Describe briefly the working principle of a Capacitive Transducer and also explain the measurement of linear displacement in a surface using the change in the area of plates.	08	2	1
	Discuss the specifications of sensors and their importance in metrology. Elucidate various types of sensors employed in measurement procedures.	08	3	2
	Explain the Velocity and motion sensors and their application in metrology for the detection of infrared radiation and temperature variations.	08	2	2
	OR			
	Describe the factors influencing the selection of sensors for a particular application. How can one ensure the compatibility between sensors and data acquisition systems?	08	3	2
	Compare and contrast position sensors and proximity sensors, highlighting their working principles and applications.	08	3	2

5	a	In a limit system the following limits are specified to give a clearance fit between a shaft and a hole: *Shaft* $30\ ^{-0.005}_{-0.018}\ mm\phi$ *Hole* $30\ ^{+0.020}_{-0.000}\ mm\phi$ Determine: i) Basic size ii) Shaft and hole tolerances iii) The shaft and hole limits iv) The maximum and minimum clearance.	
	b	Define Fit. Explain briefly the various types of fits with examples.	
		OR	
6	a	Enumerate on the concept: i) Interchangeability. ii) Hole basis systems.	
	b	Design the 'General' type *'GO'* and *'NOGO'* gauges for the components having $50H\ 8\ c\ 8$ fit. Given the following with usual notations. i) Upper deviation for Shaft = $(95 + 0.8D)$ microns ii) $50mm$ falls in the diameter step of $50 - 80mm$ iii) $IT8 = 25i$ Take wear allowance as 10% of the gauge Tolerance. Also mention the type of fit. Represent the dimensions graphically.	08
7	a	Enumerate on the working of a floating carriage micrometer and explain how the minor and effective diameter of a screw thread can be measured.	
	b	With the help of a neat sketch briefly explain the method of determining the chordal thickness of a gear tooth using gear tooth Vernier Calliper.	08 08
		OR	
8	a	Explain the constant chord method for measuring the gear tooth thickness.	08
	b	Describe briefly the working principle of a Tool Makers Microscope and explain how the minor and effective diameter of a screw thread can be measured.	08
9	a	Enumerate on the different types of main structure of a Coordinate Measuring Machine with neat sketches, and list out the industrial applications of a *CMM*.	08
	b	Describe briefly the working of a Michelson Interferometer and list out its applications.	08
		OR	
10	a	Describe the role of *CMM* in Computer Aided Manufacturing and explain briefly the advantages of *CMM*.	08
	b	Explain the basic principle of *LASER*. List the various uses of LASER in metrology and explain any two in detail.	08